

Punishment or Pruning? Diagnosing Algorithmic Collusion among Reinforcement-Learning Traders

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Abstract

Reinforcement-learning traders often settle on supracompetitive outcomes without communicating: in Kyle markets, Q-learning informed traders trade far less than competition implies, earning near-cartel profits and making prices less informative. Whether this is *collusion*, an outcome sustained by the threat of punishment, or a *learning bias* that prunes aggressive actions matters for policy: the first calls for limits on transparency, the second for standards on algorithm design. We develop diagnostics that separate the two: a paired rival-deviation test, a noise-shock test, a no-memory control, learning-rate sensitivity, and a new *myopic placebo* that retrains the same learners with no concern for the future and therefore no reason to punish. We show analytically that in the standard Kyle market a one-period deviation moves order flow by only $(I - 1)/(2I)$ noise standard deviations per unit of the value signal, whatever the volume of noise trading, while with a large mass of information-insensitive investors the ratio is in the hundreds, which explains why punishment-based collusion has been reported only in that regime. In simulations with hundreds of independent markets per configuration, Q-learners in the standard Kyle market close on average 0.86 of the gap between competition and the cartel, yet rivals never punish deviations, even with perfect monitoring of each other's orders, and deviating always pays. The myopic placebo, which cannot collude, trades even less (index 1.24, beyond the cartel), and a single trader with no rival under-trades by up to 57%, a shortfall proportional to the square root of the step size. Interventions confirm the diagnosis: reducing transparency leaves trading unchanged (0.90), an uninformative random memory of the same size reproduces the outcome (0.84), and counterfactual updating restores competition (-0.10). In the regime with information-insensitive investors, where deviations are hundreds of noise standard deviations large, separate-table Q-learners still do not punish, and with a shared Q-table the myopic placebo reproduces the threshold-shaped response to price shocks that has been read as a price trigger. We release a validated, CPU-only testbed with exact benchmarks, tabular and deep RL agents, and all diagnostics.

1 Introduction

Trading is increasingly delegated to algorithms that learn from their own profits. A growing literature shows that such algorithms can arrive at *supracompetitive* outcomes without communicating: reinforcement-learning (RL) pricing agents settle on prices above the competitive level [5, 18], algorithmic market makers quote wider spreads [8, 9], and RL informed speculators trade less aggressively than competition would imply, earning collusive profits and degrading price informativeness [12]. For regulators the finding is alarming, because tacit collusion by algorithms falls outside the reach of rules written around agreements among humans [15, 17].

A supracompetitive outcome, however, is not the same thing as collusion. In the game-theoretic sense, collusion is an outcome sustained by the threat of punishment: a trader who deviates to grab

a larger share of the rent is punished by its rivals, so deviating does not pay. The same low-intensity outcome can instead come from a learning bias: an algorithm that systematically undervalues aggressive actions will trade conservatively even when no rival could ever punish it. Dou et al. [12] name both channels in the trading context (“price-trigger” collusion and “over-pruning” bias), and a parallel debate in industrial organization asks whether Q-learning collusion is genuine or an artifact of exploration and learning design [1, 2, 4, 6, 10]. The distinction is not academic. If algorithms punish, transparency matters, because monitoring is what makes punishment possible, and the natural remedies target information: reporting delays, reduced order-flow disclosure. If the outcome is a learning bias, transparency is irrelevant and the remedies target algorithm design.

This paper asks how to tell the two apart, and what the answer is for RL informed traders in the canonical Kyle [19] market and in the Dou et al. [12] extension with information-insensitive investors.

Contributions.

1. **Diagnostics that can fail.** We combine a paired *rival-deviation* test (one trader is forced to play its one-period best response; we measure the rivals’ reaction under common random numbers) and the Dou et al. [12] *noise-shock* test with three placebos and falsification tests: a *myopic placebo* (the same learners with $\gamma = 0$, which have no reason to punish or fear punishment), an *uninformative memory* (a random state of the same size as an informative one), and *passive traders* that move the collusive outcome above the Nash one, so that collusion and bias predict opposite directions.
2. **A detectability result.** In the standard Kyle market a one-period deviation from the collusive intensity moves order flow by only $(I - 1)/(2I)$ noise standard deviations per unit of v/σ_v , whatever the volume of noise trading (Proposition 2). With a large mass of information-insensitive investors the ratio becomes $\xi\sigma_v(I - 1)/(4I\sigma_u)$, about 625 in the Dou et al. [12] calibration. This explains from first principles why punishment-based collusion has been reported only in that regime.
3. **The bias, isolated.** A single informed trader with no rival and a fixed price rule under-trades by up to 57% under standard Q-learning, by an amount proportional to $\sqrt{\alpha}$ ($R^2 = 0.994$), while counterfactual updating learns the optimum.
4. **Low trading without collusion.** In the standard Kyle market, forward-looking Q-learners close on average 0.86 of the gap between competition and the cartel, yet rivals never punish deviations, even with perfect monitoring, and deviating pays. The myopic placebo trades even less ($\Delta = 1.24$, beyond the cartel); an uninformative memory reproduces the outcome (0.84); and when passive traders put the cartel *above* Nash, learners still trade half their Nash intensity and earn 30% less than in equilibrium.
5. **What fixes it.** Blurring the order flow traders observe changes nothing ($\Delta = 0.90$), while counterfactual updating, a change to the learning rule alone, restores competition ($\Delta = -0.10$). In the Dou et al. [12] regime, where deviations are glaring, our separate-table learners match the independent replication’s collusion level (0.31) and still do not punish. With a shared Q-table, a *myopic* learner reproduces both the high collusion index (0.86) and the threshold-shaped response to noise shocks that has been read as a price trigger, so that signature does not identify collusion without a placebo.

6. **An open testbed.** A batched, CPU-only simulator with exact benchmarks, tabular Q-learning, DQN and PPO agents, all diagnostics, and a test suite that validates the economics and the diagnostics against scripted traders whose behaviour is known.

Why this matters for policy. The two channels call for different interventions. Our results suggest that in markets like the standard Kyle setting, the supracompetitive outcomes produced by off-the-shelf Q-learning are properties of the learning procedure, and would respond to algorithm-design standards (step sizes, exploration) rather than to limits on transparency. Where deviations are highly detectable, as with a large mass of information-insensitive investors, the diagnostics must be applied with a placebo, because a reaction to a price shock is not by itself evidence of punishment.

2 Related Work

Algorithmic collusion. Calvano et al. [5] show that Q-learning agents in a repeated Bertrand game learn supracompetitive prices sustained by punishment, documented through impulse responses to a forced deviation; Klein [18] obtains similar results under sequential pricing. A second wave questions how these outcomes arise. Asker et al. [2, 3] show that the outcome depends on algorithm design: asynchronous updating (learning only from the action taken) yields near-monopoly prices, while synchronous, counterfactual updating yields competitive ones. Abada and Lambin [1] attribute seemingly collusive outcomes to imperfect exploration and show that a regulator can restore competition by intervening in the learning process. Banchio and Mantegazza [4] show that collusion can arise from spontaneous coupling of the learning dynamics rather than from punishment strategies. den Boer et al. [10] argue that the collusive equilibria of Q-learning are reached only on time scales irrelevant in practice and require implausibly synchronized algorithms. Calvano et al. [6] discuss when algorithmic collusion is genuine or spurious. Deep RL agents [11] and large language models [16] can also price above competitive levels. We contribute a placebo-based diagnostic (the myopic learner) and a detectability analysis to this debate, in a financial market with adverse selection.

Algorithms in financial markets. Colliard et al. [8] study Q-learning market makers and find mark-ups over the competitive price driven by limited experimentation and noisy feedback, the same mechanism we find for informed traders. Cartea et al. [7] show that the tick size bounds the excess rents of learning liquidity providers, and Cont and Xiong [9] study tacit collusion among market-making algorithms in dealer markets. Dou et al. [12] extend Kyle [19] to several informed speculators, information-insensitive investors, and a market maker who trades off pricing error against inventory. They show that Q-learning speculators sustain collusive profits and reduce price informativeness, and separate price-trigger collusion from over-pruning bias with noise-shock impulse responses, finding the former only when information-insensitive investors are present. An independent reimplementation [13] reproduces both regimes and finds that the level of collusion depends on the update protocol. Our paper adds the rival-deviation test, the myopic placebo and the learning-rate sensitivity analysis, explains the regime difference analytically, and provides a validated open testbed.

Estimation bias in Q-learning. The max operator makes Q-learning over-estimate action values under noise [24, 25]. The bias relevant here runs the other way for rarely chosen actions: with a constant step size and ϵ -greedy exploration, an action whose estimate falls below the greedy action by chance is rarely updated again, so pessimistic errors persist while optimistic ones are corrected.

Actions with noisier payoffs are more exposed. Decreasing step sizes that satisfy the Robbins and Monro [21] conditions remove the persistent component asymptotically [14].

3 Model

3.1 The repeated market

Time is discrete, $t = 1, 2, \dots$. Each period a fresh asset value v_t is drawn i.i.d. with mean $\bar{v}$ (normalised to 0) and variance σ_v^2 . $I \geq 2$ learning informed traders observe v_t and submit market orders $x_{i,t}$. Noise traders submit $u_t \sim \mathcal{N}(0, \sigma_u^2)$, and $P \geq 0$ optional *passive* informed traders play a fixed Nash strategy. Following Dou et al. [12], a mass $\xi \geq 0$ of information-insensitive investors demands $z_t = -\xi(p_t - \bar{v})$. A market maker observes aggregate informed and noise flow $y_t = \sum_i x_{i,t} + u_t$ and sets

$$p_t = \arg \min_p \theta (p - \mathbb{E}[v_t | y_t])^2 + (y_t + z_t)^2 \implies p_t = \bar{v} + \lambda y_t, \quad \lambda = \frac{\theta \lambda_B + \xi}{\theta + \xi^2}, \quad (1)$$

where $\lambda_B = \text{Cov}(v, y) / \text{Var}(y)$ is the Bayesian price impact. With $\xi = 0$ this is the Kyle [19] market maker. Trader i earns $\pi_{i,t} = (v_t - p_t) x_{i,t}$. The value is revealed at the end of the period, and the one-period game repeats.

3.2 Benchmarks

With linear strategies $x_i = \beta_i v$, profits depend only on second moments, so the benchmarks below hold for any value distribution as long as pricing is linear. Let $n = I + P$.

Proposition 1 (Nash and collusive benchmarks). *Let $\xi = 0$. The symmetric Nash equilibrium of the stage game has $\beta^N = \sigma_u / (\sqrt{n} \sigma_v)$ and $\lambda^N = \sqrt{n} \sigma_v / ((n+1) \sigma_u)$. If the learners jointly maximise their profit against a market maker who prices their joint strategy rationally, while passive traders keep playing β^N (aggregate $B_P = P \beta^N$), the learners' aggregate intensity is*

$$B_L^C = \sqrt{B_P^2 + \sigma_u^2 / \sigma_v^2}.$$

For $\xi > 0$, the Nash intensity solves $\beta = 1 / ((n+1) \lambda(n\beta))$, which has a unique root, and the collusive intensity maximises $B_L \sigma_v^2 (1 - \lambda(B_L + B_P)(B_L + B_P))$.

Collusion lowers aggregate intensity ($B_L^C < I \beta^N$ for $P = 0$), raises learners' profit, and reduces price informativeness $\mathcal{I} = \text{Corr}(p, v)^2 = B^2 \sigma_v^2 / (B^2 \sigma_v^2 + \sigma_u^2)$. Following Calvano et al. [5] we report any outcome metric m as a *collusion index*

$$\Delta_m = \frac{m - m^N}{m^C - m^N},$$

so $\Delta = 0$ is competition and $\Delta = 1$ is the cartel. We use aggregate intensity (the estimated slope of learners' total order on v), informativeness, and profit. Because the Nash–cartel profit gap is only 6% at $I = 2$, intensity and informativeness are the more precise measures.

Remark 1 (Passive traders can remove the gain from collusion). With $\xi = 0$ and $\sigma_v = \sigma_u$, $B_L^C = \sqrt{P^2 / n + 1}$ and $I \beta^N = I / \sqrt{n}$, which coincide exactly when $I = P + 1$. In that case the learners have nothing to gain from colluding and Δ is undefined. When $P > I - 1$ the order reverses: the collusive aggregate *exceeds* the Nash aggregate, because extra informed flow lowers the market maker's price impact, so collusion would mean trading more. This makes passive traders a useful falsification test (Section 5.4).

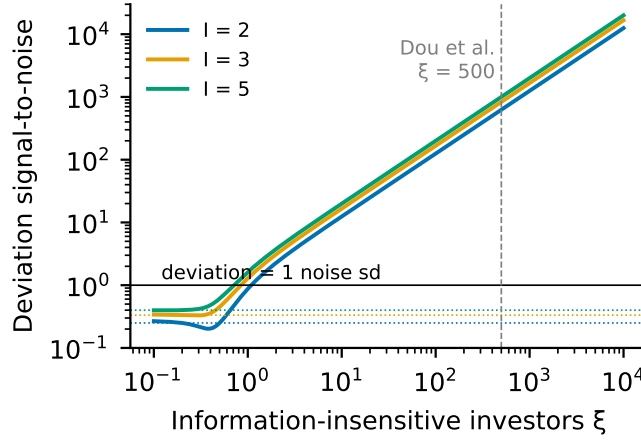


Figure 1: Signal-to-noise ratio of a one-period best-response deviation from the collusive intensity (order-flow change per unit of v/σ_v , in noise standard deviations), as a function of the mass of information-insensitive investors ($\sigma_v = 1$, $\sigma_u = 0.1$, $\theta = 0.1$). Dotted lines: the $\xi = 0$ values $(I - 1)/(2I)$. Deviations become visible above noise only once ξ exceeds about one; at the Dou et al. [12] calibration they are several hundred noise standard deviations large.

3.3 How detectable is a deviation?

Punishment requires monitoring. The relevant quantity is how much a deviation moves what traders observe relative to noise. Suppose rivals play the collusive intensity and the market maker prices at λ^C . A trader's myopic best response is $\beta^d = (1 - \lambda^C(I - 1)\beta^C)/(2\lambda^C)$, and a one-period deviation moves aggregate flow by $(\beta^d - \beta^C)v$ against noise of standard deviation σ_u .

Proposition 2 (Deviation signal-to-noise). *Let $P = 0$ and define $\text{SNR} = (\beta^d - \beta^C)\sigma_v/\sigma_u$, the flow change of a best-response deviation per unit of v/σ_v , in noise standard deviations.*

1. If $\xi = 0$, $\text{SNR} = (I - 1)/(2I) < 1/2$ for all I , independently of σ_u and σ_v .
2. As $\xi \rightarrow \infty$ with θ fixed, $\text{SNR} = \xi\sigma_v(I - 1)/(4I\sigma_u)(1 + o(1))$.

The proof is in Appendix A; Figure 1 plots the exact ratio. In the standard Kyle market every intensity scales with σ_u/σ_v , so changing noise volume rescales deviations and noise alike and cannot make monitoring easier. A single deviation is less than half a noise standard deviation even at $|v| = \sigma_v$, so a trigger strategy tuned to catch it would mostly fire on noise. With information-insensitive investors, price impact is pinned near $1/\xi$, intensities scale with ξ , and deviations become enormous relative to noise: in the Dou et al. [12] calibration ($\sigma_v = 1$, $\sigma_u = 0.1$, $\xi = 500$, $\theta = 0.1$, $I = 2$) the ratio is about 625. Detectability therefore predicts exactly the regime split that Dou et al. [12] report: over-pruning only at $\xi = 0$, price triggers at $\xi = 500$.

4 Learning Traders and Diagnostics

4.1 Learning algorithms

Tabular Q-learning. Each trader in each market owns a Q-table indexed by a memory state s (what it remembers about the previous period), the current value v , and an order from a finite grid.

It plays ε -greedy with $\varepsilon_t = e^{-\beta t}$ and updates

$$Q(s_t, v_t, x_t) \leftarrow Q(s_t, v_t, x_t) + \alpha_t \left[\pi_t + \gamma \max_{x'} Q(s_{t+1}, v_{t+1}, x') - Q(s_t, v_t, x_t) \right].$$

Following Calvano et al. [5] and Dou et al. [12], Q is initialised to the discounted payoff of each order against uniformly random rivals, and the step size is constant ($\alpha_t = \alpha$) unless stated otherwise. As an alternative we use per-entry decaying step sizes $\alpha_n = \alpha(1 + n/\kappa)^{-0.7}$, where n counts updates of the entry, which satisfy the Robbins and Monro [21] conditions.

Deep RL. Double DQN [20, 26] and PPO with generalised advantage estimation [22, 23] use the same information as continuous features (current value, previous value, and the memory signal), the same order grid, and one independent network per trader per market.

Memory. We vary what traders remember from the previous period: *none* (no memory; punishment is impossible); *flow* (binned aggregate order flow); *residual* (previous value and binned flow net of own order, i.e. what everyone else and noise traders did); *orders* (previous value and the rival’s exact order: perfect monitoring); and *price* (previous value and binned price surprise, the state used by Dou et al. [12]).

Market maker. The market maker re-estimates λ_B from exponentially weighted moments of (v, y) with a half-life of 2,000 periods and prices with (1), so it responds to the learners’ strategies as they evolve.

4.2 Diagnostics

After training, exploration and learning are switched off and greedy play is evaluated for 20,000 periods while the market maker keeps adapting.

D1: rival-deviation test. From a point on the greedy path we clone each market. In the clone, trader 1 plays its myopic best response for one period; afterwards everyone follows the learned strategies in both copies, which share every value and noise draw. We report the rivals’ change in trading intensity at each lag k , $\Delta\beta_k^{\text{rival}} = \mathbb{E}[v_k \Delta x_k^{\text{rival}}] / \text{Var}(v)$, and the deviator’s discounted gain over the window. Punishment means $\Delta\beta_1^{\text{rival}} > 0$ and a negative gain. This is the test Calvano et al. [5] use for pricing algorithms, adapted to a stochastic market with paired simulations.

D2: noise-shock test. Following Dou et al. [12], the clone receives an extra noise-trader order in the direction of v , moving the price the way a rival’s over-trading would although nobody deviated. Under price-trigger strategies traders trade harder next period; under over-pruning they do not react. We size shocks in *deviation units*: 1 is the flow change of a best-response deviation at $|v| = \mathbb{E}|v|$, so shocks mean the same thing in both regimes (Proposition 2).

D3: myopic placebo. We retrain the same learners with $\gamma = 0$. A myopic trader values only today’s profit, so it gains nothing from punishing and fears no punishment. Its Q-values estimate immediate expected profit, and any systematic reaction it shows to deviations or shocks cannot be a punishment strategy. If myopic learners reproduce a “collusive” outcome or signature, that outcome or signature does not identify collusion.

D4: no-memory control. With memory *none*, traders cannot condition on the past, so no punishment strategy exists; any supracompetitive outcome is a learning bias.

D5: learning-rate sensitivity. A strategic equilibrium is a property of the game; a bias of the learning procedure should scale with the step size and weaken under decaying step sizes. We vary the constant α and compare with decaying step sizes.

Convergence. Constant-step Q-learning with noisy payoffs keeps estimates moving, so greedy strategies need not settle. We report the share of greedy table entries that changed over the last 10^5 training periods and the mean on-path change in grid steps.

4.3 Experimental design and implementation

Unless noted, $I = 2$, $P = 0$, $\sigma_v = 1$, $\gamma = 0.95$, and each configuration is run for 100–200 independent markets. Standard-Kyle experiments ($\xi = 0$) use $\sigma_u = 1$, 5 values and a 31-point order grid spanning ± 1.5 times the largest Nash order; results in this regime are invariant to σ_u (Proposition 2). The Dou et al. [12] replication uses their calibration ($\sigma_u = 0.1$, $\theta = 0.1$, $\alpha = 0.05$), 10 values, a 15-point per-value grid spanning the cartel-to-Nash range plus 10%, and the price state. Confidence intervals are 95% across markets. The simulator advances all markets of a configuration as one batched array operation, which makes each experiment a matter of minutes to hours on a 4-core CPU. The code, configurations and a 68-test suite are released with the paper; the tests check the benchmarks against numerical optimisation, reproduce the Nash and cartel outcomes when the corresponding strategies are played, and verify that D1 and D2 report a reaction for scripted punishing traders and none for scripted memoryless ones.

5 Results

5.1 Pruning without a game

We start with the simplest possible setting: one informed trader ($I = 1$), a price rule frozen at the monopoly value, no memory, and $\gamma = 0$. There is no rival, no future, and nothing strategic; the optimal order is $v/(2\lambda)$. Figure 2 shows the learned intensity relative to the optimum across 100 independent runs per step size. Standard Q-learning under-trades, reaching only 0.87 of the optimal intensity at $\alpha = 0.02$ and 0.43 at $\alpha = 0.4$. The shortfall follows a square-root law, $1 - \hat{\beta}/\beta^* \approx 0.91\sqrt{\alpha}$ ($R^2 = 0.994$). Counterfactual updating, which credits every order with the profit it would have earned, learns the optimum at every step size (1.00–1.01).

The mechanism is the selection effect described in Section 2. With a constant step size, an action’s estimate is an exponentially weighted average of its recent payoffs and fluctuates with standard deviation proportional to $\sqrt{\alpha}$ times the payoff noise. In a Kyle market the payoff noise of an order is $\lambda\sigma_u|x|$, so larger orders have noisier estimates. An order whose estimate dips below the greedy order is updated only during exploration, so the dip persists; an order whose estimate rises is chosen and quickly corrected. The asymmetry acts like a penalty proportional to $\sqrt{\alpha}\lambda\sigma_u|x|$, which shifts the learned order toward zero by an amount proportional to $\sqrt{\alpha}$ (Appendix B gives a heuristic derivation). Counterfactual updates remove the selection, because every order is updated every period.

The bias also depends on the discount factor (Figure 2b). At $\alpha = 0.15$ the learned intensity rises from 0.64 of the optimum at $\gamma = 0$ to 0.73, 0.82 and 0.89 at $\gamma = 0.5, 0.8, 0.95$. With $\gamma > 0$ every update target also contains the continuation value $\gamma \max_{x'} Q(s', v', x')$, whose noise comes from the

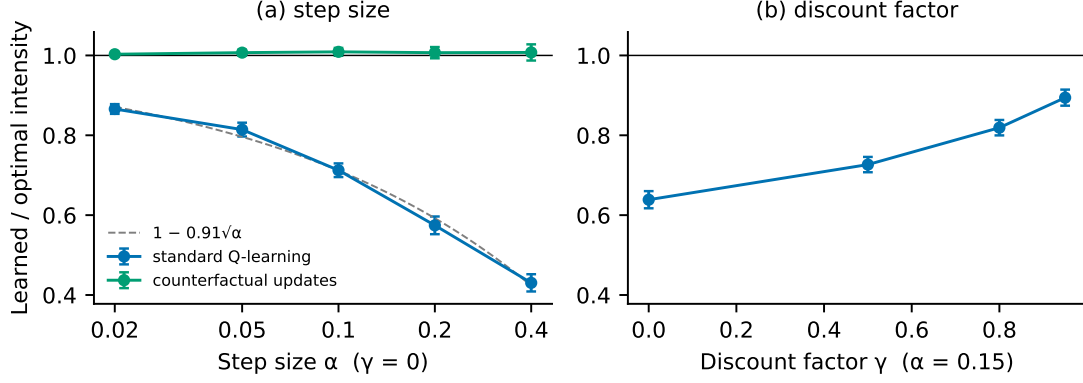


Figure 2: Pruning without a game. One informed trader facing a fixed price rule ($I = 1$, no memory, 3×10^5 periods, 100 runs per point, 95% CIs). (a) With $\gamma = 0$, standard Q-learning under-trades by about $0.91\sqrt{\alpha}$ (dashed); counterfactual updates learn the optimal order at every step size. (b) At $\alpha = 0.15$ the shortfall shrinks as the discount factor rises.

next value draw and is the same whatever order was submitted. This adds action-independent noise that dilutes the size-dependent payoff noise behind the selection effect. Myopic learners therefore prune more than forward-looking ones, for reasons that have nothing to do with strategy.

This is the over-pruning bias of Dou et al. [12], here isolated from any interaction: it alone produces trading intensities 11–57% below the optimum. Any test for collusion in a market of such learners must therefore separate this bias from strategic behaviour.

5.2 The standard Kyle market: low trading without punishment

Table 1 and Figure 3 report outcomes for two learners in the standard Kyle market ($\xi = 0$) with a constant step size $\alpha = 0.15$, 200 markets per seed and three seeds per condition.

Outcomes. Forward-looking learners that remember what their rivals and the noise traders did (*residual* memory) trade far below Nash: their aggregate intensity covers 0.86 of the distance from competition to the cartel (seed means $0.85 - 0.88$), informativeness 0.93, and profit 0.46. With perfect monitoring of each other’s orders, the index is 0.82. Without memory, where punishment is impossible, learners still close 0.24 of the gap. Read conventionally, these numbers describe substantial collusion.

The myopic placebo. Retraining the residual-memory learners with $\gamma = 0$, which removes any reason to punish or to fear punishment, produces an intensity index of 1.24 and an informativeness index of 1.46: myopic learners trade *less than the cartel* and make prices less informative than full collusion would. They do not, however, earn collusive profits (Δ profit = -0.01), because trading below the cartel level gives up profit. A learner that cannot collude thus scores higher on the intensity- and informativeness-based indices than learners that might. These indices measure how little algorithms trade, not why, and cannot by themselves identify collusion.

Punishment tests. The rival-deviation test is decisive. Figure 4 shows that after a forced one-period best-response deviation, rivals’ trading intensity is unchanged at every lag in every condition: the reaction at the first lag is -0.000 ± 0.003 for residual memory and 0.008 ± 0.008

Condition	Δ intensity	Δ inform.	Δ profit	Rival $\Delta\beta_1$	$\Delta\beta_1/\Delta\beta_0^{\text{dev}}$	Deviator gain	Markets
Remembers rivals ($\gamma = 0.95$)	0.86 ± 0.01	0.93 ± 0.01	0.46 ± 0.02	-0.000 ± 0.003	-0.00	0.053 ± 0.003	600
Remembers rivals, myopic ($\gamma = 0$)	1.24 ± 0.00	1.46 ± 0.01	-0.01 ± 0.01	0.002 ± 0.003	0.00	0.124 ± 0.004	600
No memory ($\gamma = 0.95$)	0.24 ± 0.02	0.26 ± 0.02	0.16 ± 0.04	0.000 ± 0.000	0.00	0.018 ± 0.001	600
No memory, myopic ($\gamma = 0$)	1.27 ± 0.03	1.49 ± 0.03	-0.03 ± 0.05	0.000 ± 0.000	0.00	0.115 ± 0.003	600
Perfect monitoring ($\gamma = 0.95$)	0.82 ± 0.01	0.90 ± 0.01	0.39 ± 0.03	0.008 ± 0.008	0.03	0.058 ± 0.007	200

Table 1: Standard Kyle market ($\xi = 0$, $I = 2$, constant $\alpha = 0.15$). Δ : collusion indices (0 = Nash, 1 = cartel) pooled over markets and seeds. Rival $\Delta\beta_1$: change in rivals’ trading intensity one period after a forced best-response deviation; next column: the same as a fraction of the deviation. Deviator gain: discounted profit gain from deviating over 16 periods. 95% confidence intervals.

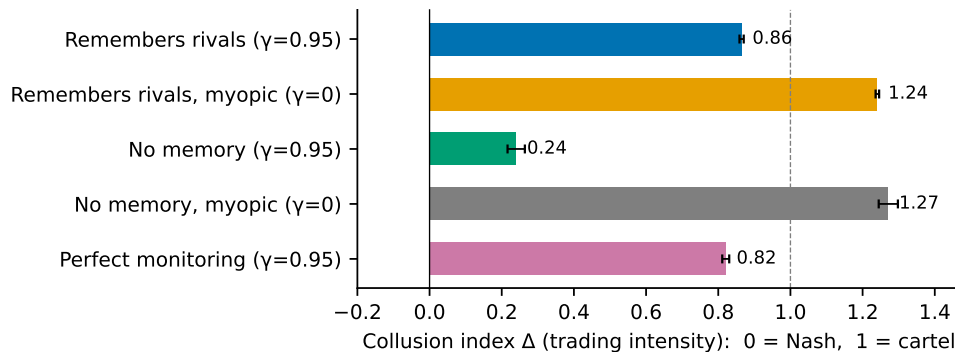


Figure 3: Collusion index (trading intensity) by condition. The myopic placebo, which cannot collude, scores beyond the cartel.

(about 3% of the deviation itself) with perfect monitoring. Deviating pays: the deviator’s discounted gain over the following 16 periods is 0.053 ± 0.003 with residual memory and 0.058 ± 0.007 with perfect monitoring, essentially all of it earned in the deviation period. No condition shows the punishment-then-reversion pattern that identifies collusion in Calvano et al. [5]. The outcome is supracompetitive but not self-enforcing, which is the signature of pruning.

This matches Proposition 2: at $\xi = 0$ a best-response deviation is about a quarter of a noise standard deviation per unit of v/σ_v , so a trigger on observed flow would fire mostly on noise. It also matches the classification of Dou et al. [12], who attribute collusion at $\xi = 0$ to over-pruning. What our diagnostics add is that even perfect monitoring, which removes the detectability problem entirely, does not lead Q-learners to discover punishment.

Convergence. With a constant step size and noisy payoffs, greedy strategies keep changing among neighbouring orders: on-path orders moved by 1.38 grid steps on average over the last 10^5 training periods (residual memory). Outcomes are nevertheless stable: seed-to-seed variation in Δ is a few hundredths, and the residual-memory results are unchanged between 6×10^6 and 1.5×10^7 training periods (Appendix Table 6).

5.3 The Dou et al. regime: detectable deviations, still no punishment

We next replicate the Dou et al. [12] setting: their calibration ($\sigma_u = 0.1$, $\theta = 0.1$, $\alpha = 0.05$), 10 values, a 15-point order grid per value spanning the cartel-to-Nash range plus 10%, and the lagged price and value as state, at $\xi = 0$ and at $\xi = 500$, where Proposition 2 puts a best-response deviation at about 625 noise standard deviations. Each trader has its own Q-table, as in the paper’s

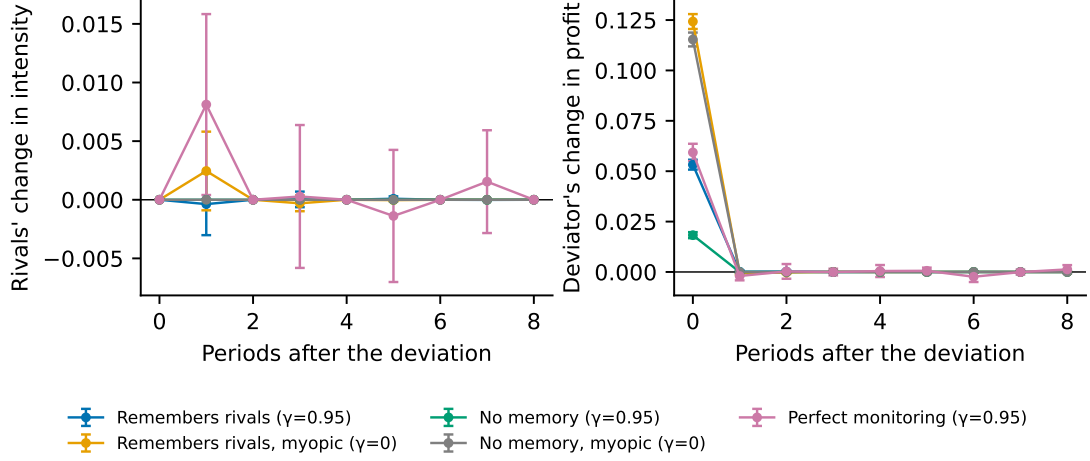


Figure 4: Rival-deviation test. Left: change in rivals’ trading intensity after trader 1 plays its one-period best response at lag 0. Right: the deviator’s change in profit. Paired simulations with common random numbers; 95% confidence intervals.

description and the primary specification of Esquinas Coves [13]. Table 3 and Figure 6 report the results.

Levels. Forward-looking learners with price memory reach $\Delta = 0.48$ at $\xi = 0$ and 0.31 at $\xi = 500$. The $\xi = 500$ value matches the 0.31 that Esquinas Coves [13] report for separate Q-tables in this regime, which supports the fidelity of our implementation. Our $\xi = 0$ value is below their 0.72, which we attribute to training horizon: their runs last 5×10^8 periods, ours 1.5×10^7 .

No punishment, even when deviations are glaring. In both regimes the learned strategies ignore noise shocks: a shock as large as a full best-response deviation (515 noise standard deviations at $\xi = 500$) changes next-period trading intensity by less than 1% of β^N , with confidence intervals covering zero, and forced deviations draw no reaction from rivals and pay. Detectability is necessary for punishment but not sufficient: separate-table Q-learners do not discover trigger strategies within our horizon even when deviations are hundreds of standard deviations large.

The placebo at $\xi = 0$. As in the standard Kyle experiments, myopic learners under-trade more than forward-looking ones at $\xi = 0$ (0.63 versus 0.48). At $\xi = 500$ the order reverses (0.11 versus 0.31), but without any reaction to shocks or deviations in either case, so the difference cannot be attributed to punishment.

Shared Q-tables and the placebo. Esquinas Coves [13] find that the price-trigger signature at $\xi = 500$ is sharpest when both traders read and update a single *shared* Q-table, and that this protocol also yields much higher collusion indices (0.87–0.92). A shared table is one learner choosing for several traders, so we apply the same placebo to it. Table 2 and Figure 5 show the results. Sharing raises the index for forward-looking learners to 0.68 ($\xi = 0$) and 0.71 ($\xi = 500$), and for myopic learners to 0.86 in both regimes, the level reported for shared tables in the replication, reached by a learner that cannot collude.

ξ	Shared Q-table	Δ intensity	Shock response (% of β^N)			Rival reaction (% of β^N)
			0.05	0.25	1	
0	$\gamma = 0.95$	0.68 ± 0.00	0.12 ± 0.13	0.15 ± 0.35	-0.06 ± 0.68	0.10 ± 0.51
0	$\gamma = 0$ (myopic placebo)	0.86 ± 0.00	-0.05 ± 0.11	0.04 ± 0.22	0.53 ± 0.59	0.07 ± 0.45
500	$\gamma = 0.95$	0.71 ± 0.01	0.12 ± 0.25	0.14 ± 0.38	0.24 ± 0.50	0.08 ± 0.20
500	$\gamma = 0$ (myopic placebo)	0.86 ± 0.00	-0.00 ± 0.03	0.75 ± 0.16	3.34 ± 0.32	0.15 ± 0.09

Table 2: Shared Q-tables in the Dou et al. [12] calibration (1.5×10^7 periods, 100 markets). Shock response: change in per-trader intensity one period after a noise shock of the given size in deviation units, as a percentage of β^N . Rival reaction: change in rivals’ intensity one period after a forced best-response deviation. 95% CIs.

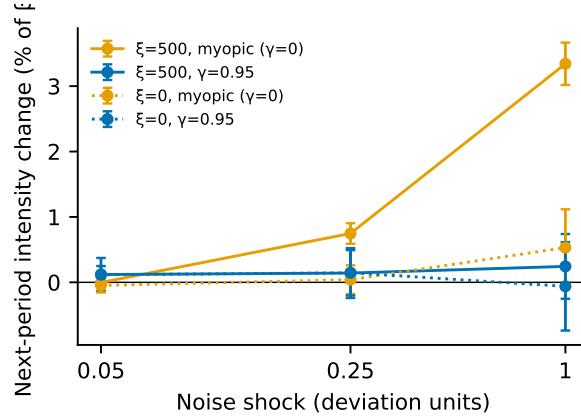


Figure 5: Noise-shock responses with shared Q-tables. At $\xi = 500$ the myopic placebo (orange, solid) ignores small shocks and reacts to large ones, the threshold pattern read as a price trigger, although with $\gamma = 0$ it has no reason to punish.

The noise-shock test then fails the placebo. At $\xi = 500$ the *myopic* shared learner ignores small shocks (-0.00% of β^N for a shock of 0.05 deviation units) but trades harder after larger ones ($0.75 \pm 0.16\%$ at 0.25 and $3.34 \pm 0.32\%$ at one deviation unit): the threshold pattern that Dou et al. [12] interpret as a price trigger. A learner with $\gamma = 0$ cannot be punishing anyone, so this response cannot be a punishment strategy. The forward-looking shared learner shows no significant response at our horizon. Appendix C shows why: greedy play visits only the two outermost price states, large shocks push the market into interior states that greedy play never visits, and there the learner’s orders are still close to their Nash-like initialisation, so it trades harder. Small shocks do not reach those states, which produces the threshold. Under pruning, the response of a Q-learner to a price shock depends on how often the resulting state has been visited, so a threshold-like reaction can arise without any strategy. The noise-shock test therefore needs the placebo to be informative: a reaction identifies punishment only if a myopic learner trained in the same way does not show it.

5.4 What changes the outcome: information or learning design?

The two channels make opposite predictions about interventions. If low trading were sustained by punishment, degrading the information traders use to monitor each other should restore competition, and the content of their memory should matter. If it is pruning, information is irrelevant and changes to the learning procedure should matter. Table 4 and Figure 7 report treatments in the

ξ	Learners	Δ intensity	Δ profit	Shock response	Rival reaction
0	$\gamma=0.95$, remembers price	0.48 ± 0.00	0.59 ± 0.04	0.005 ± 0.006	-0.007 ± 0.005
0	$\gamma=0$ (myopic placebo)	0.63 ± 0.00	0.74 ± 0.04	0.003 ± 0.005	0.003 ± 0.006
0	$\gamma=0.95$, no memory	0.37 ± 0.02	0.50 ± 0.05	0.000 ± 0.000	0.000 ± 0.000
500	$\gamma=0.95$, remembers price	0.31 ± 0.01	0.45 ± 0.02	-0.006 ± 0.005	-0.002 ± 0.004
500	$\gamma=0$ (myopic placebo)	0.11 ± 0.00	0.18 ± 0.01	-0.000 ± 0.003	-0.003 ± 0.004
500	$\gamma=0.95$, no memory	0.24 ± 0.02	0.36 ± 0.03	0.000 ± 0.000	0.000 ± 0.000

Table 3: Dou et al. [12] calibration with separate Q-tables (1.5×10^7 periods, 100 markets). Shock response: change in per-trader intensity one period after a noise shock equal to one best-response deviation, as a fraction of β^N . Rival reaction: change in rivals’ intensity one period after a forced best-response deviation, as a fraction of β^N . The informativeness index is omitted because at $\xi = 500$ the Nash and cartel informativeness both equal one to four decimals. 95% CIs.

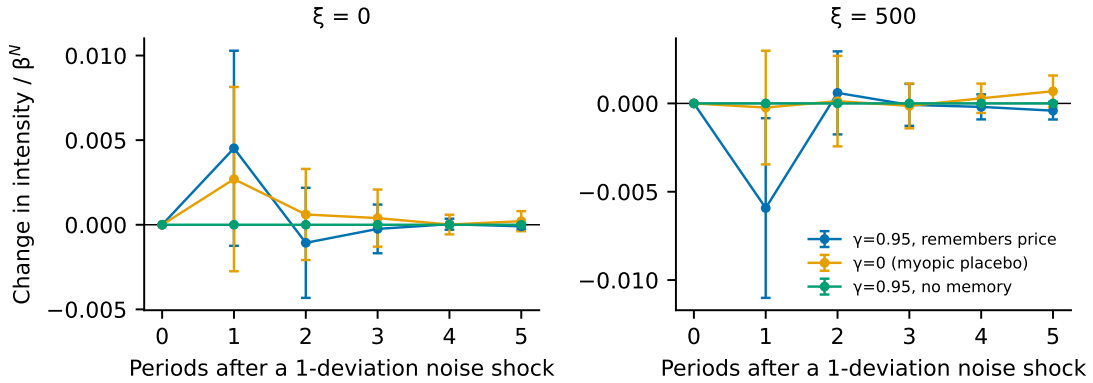


Figure 6: Noise-shock impulse responses in the Dou et al. [12] calibration. A noise-trader shock equal to one best-response deviation hits at lag 0; lines show the change in per-trader intensity relative to β^N . Under price-trigger strategies intensity would jump at lag 1; it does not.

standard Kyle market, each against the residual-memory baseline ($\Delta = 0.86$).

Information does not matter. Blurring the order flow that traders observe with noise of twice the noise-trading volume leaves the outcome unchanged ($\Delta = 0.90$). Replacing the informative memory with a *random* state that carries no information at all, but has the same number of states, reproduces it ($\Delta = 0.84$). The difference between forward-looking learners with and without memory (0.86 versus 0.24) is therefore a matter of table size: more states mean fewer updates per entry and more room for the selection effect of Section 5.1, not more information to coordinate on.

Learning design matters. Two changes to the learning rule move the outcome substantially. Counterfactual updating, which removes the selection effect, eliminates the supracompetitive outcome: forward-looking learners end at $\Delta = -0.10$ with residual memory and -0.09 without, that is, at (marginally beyond) the competitive level, with price informativeness at its Nash value (Δ informativeness -0.01). The market, the agents’ information and the exploration schedule are unchanged; only the update rule differs. Per-entry decaying step sizes change even the sign: with the schedule of Section 4, learners *over-trade* relative to Nash ($\Delta = -0.50$ with residual memory,

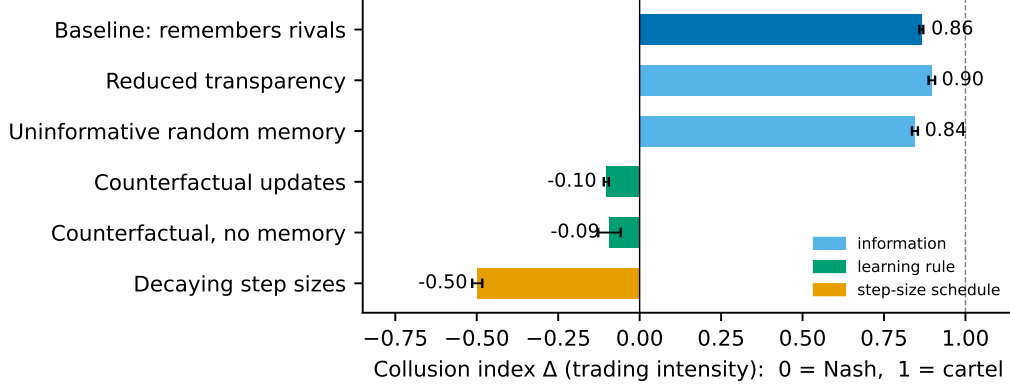


Figure 7: Interventions in the standard Kyle market. Treatments that change what traders observe leave the outcome unchanged; treatments that change how they learn move it to, or past, competition.

Treatment ($\xi = 0, I = 2$)	Δ intensity	Δ profit	Rival $\Delta\beta_1$	Deviator gain
Random memory, 35 states	0.84 ± 0.01	0.45 ± 0.03	0.000 ± 0.000	0.052 ± 0.004
Random memory, myopic	1.16 ± 0.01	0.11 ± 0.02	0.000 ± 0.000	0.107 ± 0.006
Counterfactual updates, residual memory	-0.10 ± 0.01	-0.44 ± 0.03	-0.001 ± 0.002	0.020 ± 0.003
Counterfactual updates, no memory	-0.09 ± 0.03	-0.34 ± 0.08	0.000 ± 0.000	0.015 ± 0.002
Counterfactual updates, residual, myopic	-0.10 ± 0.01	-0.44 ± 0.03	-0.001 ± 0.002	0.019 ± 0.003
Reduced transparency (disclosure noise $2\sigma_u$)	0.90 ± 0.01	0.46 ± 0.03	-0.002 ± 0.003	0.058 ± 0.004
Two passive Nash traders	-1.26 ± 0.01	-8.56 ± 0.09	0.001 ± 0.003	0.036 ± 0.003
DQN	-0.36 ± 0.07	-0.89 ± 0.13	-0.000 ± 0.004	0.026 ± 0.008
DQN, myopic	-0.17 ± 0.03	-0.46 ± 0.06	-0.001 ± 0.005	0.014 ± 0.007
PPO	-0.02 ± 0.09	-0.27 ± 0.12	0.000 ± 0.000	0.028 ± 0.008
PPO, myopic	-0.28 ± 0.05	-0.51 ± 0.10	0.000 ± 0.000	0.012 ± 0.006

Table 4: Treatments in the standard Kyle market ($\xi = 0, I = 2$, residual memory unless noted, 6×10^6 periods, 200 markets). Baseline: Δ intensity 0.86. 95% CIs.

−0.60 without), because step sizes shrink while the market maker’s price impact is still depressed by heavy exploration, freezing estimates around best responses to a market that is too cheap to trade in. These learners are not converged either (on-path orders still move by 4.74 grid steps), which underlines how strongly the learning procedure, rather than the game, shapes the outcome.

When collusion means trading more. Adding two passive informed traders who play the Nash strategy reverses the benchmarks: facing outside competition, the two learners would maximise their joint profit by trading *more* than their Nash intensity (aggregate 1.41 versus 1.00), because extra informed flow lowers the market maker’s price impact. A collusive force should push them up. Instead they trade 0.48, half their Nash intensity, moving away from the cartel and earning 30% less than in the Nash equilibrium. The learners under-trade whichever side of Nash the collusive outcome lies on, which is what a learning bias predicts and the opposite of what collusion predicts. (In the collusion index this appears as $\Delta = -1.26$, which is why Table 4 should be read with the benchmarks in mind.)

Learner (residual memory)	Δ intensity	Δ informativeness	Rival $\Delta\beta_1$	Deviator gain
Tabular Q, $\gamma = 0.95$	0.86 ± 0.01	0.93 ± 0.01	-0.000 ± 0.003	0.053 ± 0.003
Tabular Q, $\gamma = 0$	1.24 ± 0.00	1.46 ± 0.01	0.002 ± 0.003	0.124 ± 0.004
DQN, $\gamma = 0.95$	-0.36 ± 0.07	-0.17 ± 0.05	-0.000 ± 0.004	0.026 ± 0.008
DQN, $\gamma = 0$	-0.17 ± 0.03	-0.09 ± 0.02	-0.001 ± 0.005	0.014 ± 0.007
PPO, $\gamma = 0.95$	-0.02 ± 0.09	0.06 ± 0.07	0.000 ± 0.000	0.028 ± 0.008
PPO, $\gamma = 0$	-0.28 ± 0.05	-0.18 ± 0.04	0.000 ± 0.000	0.012 ± 0.006

Table 5: Learning algorithms in the standard Kyle market ($\xi = 0$, $I = 2$, residual memory). Tabular Q: 600 markets, 6×10^6 periods; DQN and PPO: 50 markets, 4×10^5 periods. 95% CIs.

5.5 Deep reinforcement learning

Table 5 repeats the standard-Kyle experiment with residual memory for Double DQN and PPO (4×10^5 periods, 50 markets each; the deep agents are far more expensive per step). Neither produces a supracompetitive outcome: PPO ends at the competitive level ($\Delta = -0.02$) and DQN slightly *over*-trades ($\Delta = -0.36$), with myopic variants close by (-0.28 and -0.17). No variant punishes deviations. This is what the mechanism of Section 5.1 predicts. Pruning requires estimates that stay frozen for orders that are rarely chosen. PPO learns from every sampled order through the policy gradient, with an entropy bonus that keeps sampling broad, and DQN’s network generalises across neighbouring orders and reuses past experience through replay, so neither leaves individual orders stranded with pessimistic estimates. We conjecture that DQN’s residual over-estimation from the max operator [24, 26], which favours orders with noisier payoffs, explains its slight over-trading. The deep-RL horizon is short and the hyperparameters standard, so we view these results as evidence that the supracompetitive outcome is specific to tabular, constant-step value learning, not as a definitive statement about deep RL in markets [cf. 11].

6 Discussion

What the diagnostics show. Four findings recur across our experiments. First, the supracompetitive outcomes of Q-learning informed traders do not rest on punishment: rivals do not react to deviations, even with perfect monitoring or when deviations are hundreds of noise standard deviations large, and deviating pays. Second, the same learning procedure under-trades with no rival at all, by an amount that grows with $\sqrt{\alpha}$ and disappears under counterfactual updates; in the market, counterfactual updating restores competition while degrading information does nothing. Third, placebos that cannot collude reproduce the conventional evidence of collusion: myopic learners score at or beyond the cartel on intensity- and informativeness-based indices, an uninformative memory reproduces the effect of an informative one, and a myopic shared-table learner shows the threshold-shaped reaction to price shocks that has been read as a price trigger. Fourth, detectability rather than learning sophistication decides whether punishment is even feasible: in the standard Kyle market a deviation is a fraction of a noise standard deviation (Proposition 2).

Implications for auditing algorithms. Our results suggest a simple protocol before treating a supracompetitive outcome as collusion: (i) run a paired deviation test and check that deviating does not pay; (ii) retrain the algorithm myopically and check that the outcome and any response to shocks disappear; (iii) replace its informative state by an uninformative one of the same size; and (iv) check whether the outcome survives a change to the update rule that removes selection effects.

Each step can fail, and a genuine punishment scheme would pass all four. None of the outcomes we study does.

Implications for policy. If supracompetitive trading is a learning bias, interventions aimed at collusion through information, such as delayed or coarser order-flow disclosure, will not work; our reduced-transparency treatment confirms this. What works is algorithm design: counterfactual (full-information) updating removes the bias. This mirrors conclusions of Asker et al. [2, 3] and Abada and Lambin [1] for pricing algorithms and extends them to informed trading. A bias is not harmless: it still lowers liquidity and price informativeness, myopic learners push trading below even the cartel level, and with passive traders it costs the learners themselves 30% of their equilibrium profit. But its remedy is a design standard, not a cartel investigation.

Limitations. Our evidence concerns tabular Q-learning with standard hyperparameters and, more briefly, DQN and PPO; learners that model their rivals may find punishment strategies these do not. Constant-step Q-learning does not converge in the strict sense of Calvano et al. [5]: greedy strategies still change on path by one to two grid steps over the last 10^5 periods, although outcomes are stable across seeds, schedules and horizons. Our training horizons (6×10^6 to 1.5×10^7 periods) are shorter than those of Dou et al. [12] and Esquinas Coves [13] (up to 5×10^{10} and 5×10^8), and the forward-looking shared-table learner, which in their runs shows a price-trigger signature at $\xi = 500$, shows none at our horizon; longer training could change this comparison, which is why the placebo, not the absence of a response, carries our argument there. The market maker learns its price impact from exponentially weighted moments rather than being fixed at the equilibrium value. Finally, the stylised market abstracts from inventory dynamics, multiple assets and long-lived information.

Conclusion. Low trading by learning algorithms is easy to produce and easy to misread. Telling collusion from learning bias requires diagnostics that can fail: a deviation test, placebos that cannot collude, and a check of whether the outcome moves with the learning rule. In the canonical market for informed trading, and in the regime where deviations are most visible, these diagnostics point to pruning, not punishment.

References

- [1] Ibrahim Abada and Xavier Lambin. Artificial intelligence: Can seemingly collusive outcomes be avoided? *Management Science*, 69(9), 2023.
- [2] John Asker, Chaim Fershtman, and Ariel Pakes. Artificial intelligence, algorithm design, and pricing. *AEA Papers and Proceedings*, 112:452–456, 2022.
- [3] John Asker, Chaim Fershtman, and Ariel Pakes. The impact of artificial intelligence design on pricing. *Journal of Economics & Management Strategy*, 33:276–304, 2024.
- [4] Martino Banchio and Giacomo Mantegazza. Adaptive algorithms and collusion via coupling. In *Proceedings of the 24th ACM Conference on Economics and Computation (EC '23)*, 2023.
- [5] Emilio Calvano, Giacomo Calzolari, Vincenzo Denicolò, and Sergio Pastorello. Artificial intelligence, algorithmic pricing, and collusion. *American Economic Review*, 110(10):3267–3297, 2020.

- [6] Emilio Calvano, Giacomo Calzolari, Vincenzo Denicolò, and Sergio Pastorello. Algorithmic collusion: Genuine or spurious? *International Journal of Industrial Organization*, 90, 2023.
- [7] Álvaro Cartea, Patrick Chang, and José Penalva. Algorithmic collusion in electronic markets: The impact of tick size. Technical Report 4105954, SSRN, 2022.
- [8] Jean-Edouard Colliard, Thierry Foucault, and Stefano Lovo. Algorithmic pricing and liquidity in securities markets. *Review of Financial Studies*, forthcoming.
- [9] Rama Cont and Wei Xiong. Dynamics of market making algorithms in dealer markets: Learning and tacit collusion. *Mathematical Finance*, 2024.
- [10] Arnoud V. den Boer, Janusz M. Meylahn, and Maarten Pieter Schinkel. Artificial collusion: Examining supracompetitive pricing by Q-learning algorithms. *Management Science*, 2026. doi: 10.1287/mnsc.2024.08557.
- [11] Shidi Deng, Maximilian Schiffer, and Martin Bichler. Algorithmic collusion in dynamic pricing with deep reinforcement learning. arXiv:2406.02437, 2024.
- [12] Winston Wei Dou, Itay Goldstein, and Yan Ji. AI-powered trading, algorithmic collusion, and price efficiency. NBER Working Paper 34054, National Bureau of Economic Research, 2025.
- [13] Clemente Esquinas Coves. Algorithmic collusion in financial markets: A revisit of Dou et al. (2025). <https://github.com/clemente-esquinas/algorithmic-collusion-kyle>, 2026.
- [14] Eyal Even-Dar and Yishay Mansour. Learning rates for Q-learning. *Journal of Machine Learning Research*, 5:1–25, 2003.
- [15] Ariel Ezrachi and Maurice E. Stucke. *Virtual Competition: The Promise and Perils of the Algorithm-Driven Economy*. Harvard University Press, 2016.
- [16] Sara Fish, Yannai A. Gonczarowski, and Ran I. Shorrer. Algorithmic collusion by large language models. arXiv:2404.00806, 2024.
- [17] Joseph E. Harrington. Developing competition law for collusion by autonomous artificial agents. *Journal of Competition Law & Economics*, 14(3):331–363, 2018.
- [18] Timo Klein. Autonomous algorithmic collusion: Q-learning under sequential pricing. *RAND Journal of Economics*, 52(3):538–558, 2021.
- [19] Albert S. Kyle. Continuous auctions and insider trading. *Econometrica*, 53(6):1315–1335, 1985.
- [20] Volodymyr Mnih et al. Human-level control through deep reinforcement learning. *Nature*, 518: 529–533, 2015.
- [21] Herbert Robbins and Sutton Monro. A stochastic approximation method. *Annals of Mathematical Statistics*, 22(3):400–407, 1951.
- [22] John Schulman, Philipp Moritz, Sergey Levine, Michael Jordan, and Pieter Abbeel. High-dimensional continuous control using generalized advantage estimation. In *International Conference on Learning Representations*, 2016.
- [23] John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms. arXiv:1707.06347, 2017.

- [24] Sebastian Thrun and Anton Schwartz. Issues in using function approximation for reinforcement learning. In *Proceedings of the Fourth Connectionist Models Summer School*, 1993.
- [25] Hado van Hasselt. Double Q-learning. In *Advances in Neural Information Processing Systems*, 2010.
- [26] Hado van Hasselt, Arthur Guez, and David Silver. Deep reinforcement learning with double Q-learning. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 2016.

A Proofs

Proposition 1. With $x_j = \beta_j v$ and $p = \lambda y$, trader i 's expected profit is $\beta_i \sigma_v^2 (1 - \lambda \sum_j \beta_j)$ (the noise term has zero mean). Taking λ as given, the first-order condition gives $\beta_i = (1 - \lambda \sum_{j \neq i} \beta_j) / (2\lambda)$, so a symmetric equilibrium has $\beta = 1 / ((n+1)\lambda)$. With $\xi = 0$, rational pricing gives $\lambda = n\beta\sigma_v^2 / (n^2\beta^2\sigma_v^2 + \sigma_u^2)$. Substituting and solving yields $\beta^N = \sigma_u / (\sqrt{n}\sigma_v)$ and $\lambda^N = \sqrt{n}\sigma_v / ((n+1)\sigma_u)$. For the collusive benchmark, total learner profit with rational pricing and aggregate $T = B_L + B_P$ is $B_L \sigma_v^2 \sigma_u^2 / (T^2 \sigma_v^2 + \sigma_u^2)$. The first-order condition in B_L is $T^2 \sigma_v^2 + \sigma_u^2 - 2B_L T \sigma_v^2 = 0$, i.e. $\sigma_v^2 (B_P^2 - B_L^2) + \sigma_u^2 = 0$, which gives B_L^C . For $\xi > 0$, $h(\beta) = (n+1)\beta\lambda(n\beta) - 1$ is strictly increasing because both $\beta\lambda_B(n\beta) = n\beta^2\sigma_v^2 / (n^2\beta^2\sigma_v^2 + \sigma_u^2)$ and $\xi\beta$ are increasing, and $h(0) = -1$, $h(\infty) > 0$, so the root is unique. $\square$

Proposition 2. (i) With $\xi = 0$ and $P = 0$, $\beta^C = \sigma_u / (I\sigma_v)$ and $\lambda^C = \sigma_v / (2\sigma_u)$. Then $\beta^d = (1 - \lambda^C(I-1)\beta^C) / (2\lambda^C) = \frac{\sigma_u}{\sigma_v} (1 - \frac{I-1}{2I}) = \frac{\sigma_u}{\sigma_v} \cdot \frac{I+1}{2I}$, so $\beta^d - \beta^C = \frac{\sigma_u}{\sigma_v} \cdot \frac{I-1}{2I}$ and $\text{SNR} = (I-1)/(2I)$. (ii) From (1), $\lambda = 1/\xi + O(\xi^{-2})$ uniformly in $\lambda_B \in [0, \sigma_v/(2\sigma_u)]$. Treating λ as constant, the collusive aggregate maximises $B(1 - \lambda B)$, so $\beta^C = 1/(2I\lambda)$, and $\beta^d = (1 - \lambda(I-1)\beta^C) / (2\lambda) = \frac{1}{2\lambda} \cdot \frac{I+1}{2I}$. Hence $\beta^d - \beta^C = \frac{1}{\lambda} \cdot \frac{I-1}{4I} = \xi \frac{I-1}{4I} (1 + o(1))$ and $\text{SNR} = \xi \sigma_v (I-1) / (4I\sigma_u) (1 + o(1))$. Numerically, the exact ratio at $I = 2$, $\sigma_v = 1$, $\sigma_u = 0.1$, $\theta = 0.1$ is 62.49, 625.00 and 6250.0 for $\xi = 50, 500, 5000$. $\square$

B Why pruning scales with $\sqrt{\alpha}$: a heuristic

Consider one trader, value $v > 0$, a fixed price rule with impact λ , and orders x with expected profit $\mu(x) = vx - \lambda x^2$ and payoff noise $\sigma(x) = \lambda \sigma_u x$. With a constant step size, the estimate of an order updated at every visit is an exponentially weighted average of past payoffs, with stationary standard deviation $s(x) = \sigma(x) \sqrt{\alpha / (2 - \alpha)} \approx \sigma(x) \sqrt{\alpha / 2}$.

Under ε -greedy play the greedy order is updated almost every period, so its estimate stays close to μ . A competing order is updated only when explored, so after an update its estimate stays frozen for a long time. If the frozen estimate is below the greedy one the order stays non-greedy until the next exploration; if it is above, the order becomes greedy, is updated repeatedly, and its estimate is pulled back to μ . The greedy order is therefore, roughly, the one that keeps winning these comparisons, which favours orders whose estimates are *reliably* high: to first order the learner behaves as if it maximised $\mu(x) - c s(x)$ for a constant $c > 0$ that depends on the exploration schedule. With $s(x) \propto \sqrt{\alpha} \lambda \sigma_u x$, the maximiser is

$$x_\alpha^* = \frac{v - c' \sqrt{\alpha} \lambda \sigma_u}{2\lambda}, \quad 1 - \frac{x_\alpha^*}{x_0^*} = c' \sqrt{\alpha} \frac{\lambda \sigma_u}{v},$$

a shortfall linear in $\sqrt{\alpha}$, as found in Figure 2a ($R^2 = 0.994$). Counterfactual updates refresh every order every period, so no order is ever frozen and the selection disappears. With $\gamma > 0$, each target

also carries continuation-value noise that does not depend on x ; this raises $s(x)$ equally for all orders and makes the comparison less sensitive to the size-dependent part, which is why the shortfall shrinks with γ (Figure 2b). We do not offer this as a proof; it is the intuition behind the selection effect, and the experiments test its predictions.

C Rare states and the shock response

D Robustness

Table 6 reports the collusion indices of forward-looking Q-learners in the standard Kyle market across training horizons, exploration schedules and seeds. Outcomes are stable once exploration is long enough; the residual-memory run with the fastest exploration decay is the exception because its larger table is under-explored (about 46 exploratory visits per entry), and it is excluded from the main results.

Configuration ($\gamma = 0.95$, $\alpha = 0.15$)	Δ intensity	Δ informativeness
Residual, 1.5×10^6 periods, $\beta = 4 \times 10^{-6}$	0.31 ± 0.02	0.68 ± 0.02
Residual, 6×10^6 , $\beta = 10^{-6}$, seed 0	0.88 ± 0.01	0.95 ± 0.01
Residual, 6×10^6 , $\beta = 10^{-6}$, seed 1	0.87 ± 0.01	0.93 ± 0.01
Residual, 6×10^6 , $\beta = 10^{-6}$, seed 2	0.85 ± 0.01	0.92 ± 0.01
Residual, 1.5×10^7 , $\beta = 4 \times 10^{-7}$	0.90 ± 0.01	0.97 ± 0.01
Flow, 1.5×10^6 , $\beta = 4 \times 10^{-6}$	0.66 ± 0.02	0.70 ± 0.02
Flow, 6×10^6 , $\beta = 10^{-6}$	0.67 ± 0.02	0.69 ± 0.02
No memory, 1.5×10^6 , $\beta = 4 \times 10^{-6}$	0.27 ± 0.04	0.28 ± 0.03
No memory, 6×10^6 , $\beta = 10^{-6}$ (3 seeds)	0.24 ± 0.02	0.26 ± 0.02
No memory, 1.5×10^7 , $\beta = 4 \times 10^{-7}$	0.27 ± 0.06	0.28 ± 0.05

Table 6: Robustness of the standard-Kyle outcomes. β is the exploration decay ($\varepsilon_t = e^{-\beta t}$). 100–200 markets per row; 95% CIs.

E Implementation details

Validation. The test suite (68 tests) checks that the benchmarks maximise the relevant objectives on fine grids, that the closed forms and the numerical solver agree at $\xi = 0$, and that the solver reproduces the Dou et al. [12] values at $\xi = 500$ ($\beta^N = 166.667$, $\beta^C = 125$, $\lambda^N = 0.002$); that simulated Nash and cartel play recover the corresponding price impact, profit and collusion index; that sessions are independent; that counterfactual profits equal realised profits for the order taken; that a resumed run reproduces an uninterrupted one exactly; and that the deviation and noise-shock tests report a reaction for scripted punishing or price-trigger traders and none for memoryless ones.

Parameter	Value
Values v	5 (standard Kyle) or 10 (Dou replication) equiprobable normal quantiles, variance rescaled to σ_v^2
Orders	31-point grid on $\pm 1.5 \beta^N \max v $ (wide); 15 per value on the cartel–Nash bracket $\pm 10\%$
Memory bins	7 flow/residual bins on ± 2 sd; 15 price-surprise bins on ± 2.5 sd
Q-learning	$\alpha = 0.15$ (0.05 in Dou replication), $\gamma = 0.95$, $\varepsilon_t = e^{-\beta t}$
Q initialisation	discounted payoff against uniformly random rivals at λ^N
Decaying step	$\alpha_n = \alpha(1 + n/50)^{-0.7}$ per entry
Market maker	EWMA moments of (v, y) , half-life 2,000 periods, starts at Nash
Evaluation	20,000 greedy periods, learning off, market maker adapting
Deviation test	40 events per market, 50 greedy periods apart, 16-period window
Noise-shock test	shocks of 0.25 and 1 deviation units, 20 events per market
DQN	2×64 MLP, Double DQN, Huber loss, Adam 10^{-3} , buffer 5×10^4 , batch 64
PPO	2×64 actor-critic, clip 0.2, GAE $\lambda = 0.95$, 4 epochs, rollout 256, entropy annealed to 0

Table 7: Default parameters.